Time in Production and the Macroeconomy

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Abstract

This paper studies how time in production shapes the propagation of macroeconomic shocks. I develop a discrete-time general-equilibrium model in which intermediate-good producers choose both labor input and the time required to produce output. Time-to-build is costly, but firms differ in their productivity of time spent in production. This generates an endogenous distribution of production durations across firms. The model shows that even one-time productivity shocks can generate persistent aggregate effects. For small shocks around the steady state, endogenous time-to-build is locally unchanged, so the economy behaves like one with exogenous production lags. For sufficiently large shocks, firms adjust production timing, reallocating output away from low-productivity dates. Anticipated and unexpected shocks have different effects because firms facing unexpected shocks may be constrained by the minimum feasible production duration. I also discuss how the time-to-build margin affects monetary-policy transmission.

1 Introduction

Recent disruptions to global production have renewed interest in how shocks propagate through the production side of the economy. A large literature studies this question through the lens of production networks, where aggregate fluctuations arise because firms rely on inputs from other firms and shocks travel through these linkages (**??????**).

This paper studies a complementary channel. Rather than focusing on the network structure of production, I focus on the fact that production takes time. Many goods are not produced instantaneously. Firms must choose when to begin production, how long to remain in production, and when output will become available for final use. This time-to-build margin changes the way shocks are transmitted across periods.

The central question is: how does time in production shape the transmission of aggregate shocks? The main answer is that even one-time productivity shocks can generate persistent effects. When production takes time, a shock at a single date affects not only firms producing for that date, but also firms that began production in earlier periods and firms that can alter production plans for future delivery. As a result, the economy inherits a dynamic propagation mechanism even in an otherwise simple environment without capital.

The paper develops a discrete-time general-equilibrium model with a representative household, a CES final-good producer, and a set of competitive intermediate-good producers. Intermediate firms differ in their productivity of time-to-build. A firm choosing a longer production period faces a physical cost of time, but may also benefit from greater time-to-build productivity. This creates an endogenous distribution of production durations across firms.

The model delivers three main insights. First, for infinitesimal shocks around the steady state, endogenous time-to-build behaves locally like exogenous time-to-build: firms do not adjust production durations, and the response of output is governed by the mass of firms scheduled to complete production at each horizon. Second, for sufficiently large anticipated shocks, firms reallocate production away from the shocked date. This dampens the direct effect of the shock but can shift production into adjacent periods. Third, unexpected shocks generate a different response because firms that are already constrained by the minimum feasible production time cannot adjust immediately. Hence, unexpected shocks have a stronger contemporaneous effect, while later firms can still adjust.

The paper also studies the role of monetary policy. Since production takes time, changes in interest rates affect the relative payoff from producing for different future dates. In a model with no time-to-build, monetary policy has no real effect through this channel. With exogenous time-to-build, monetary policy can affect output mechanically by changing the valuation of future production. With endogenous time-to-build, however, firms may adjust production timing in response to interest-rate changes, attenuating or reshaping the output response.

2 Model

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. The economy consists of a representative household, a competitive final-good producer, and a set of competitive intermediate-good producers indexed by

$$i \in \{0, \ldots, F\}.$$

There is no capital. The key friction is that intermediate goods require time to produce.

2.1 Households

The representative household chooses consumption C_t and labor supply L_t to maximize

$$\max_{\{C_t, L_t\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t \theta_t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{L_t^{1+\eta}}{1+\eta} \right),$$

subject to the period budget constraint

$$P_t C_t = W_t L_t,$$

where P_t is the final-good price and W_t is the nominal wage. The parameter $\sigma > 0$ governs the curvature of consumption utility, $\eta > 0$ governs the curvature of labor disutility, and $\chi > 0$ scales the disutility of labor.

The household first-order condition implies

$$\chi L_t^{\eta} C_t^{\sigma} = \frac{W_t}{P_t}.$$

2.2 Final-Good Production

The final good is produced competitively using intermediate goods according to the CES aggregator

$$Y_\tau = \left(\sum_{i=0}^{F} \tilde{y}_{i,\tau}^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\tilde{y}_{i,\tau}$ denotes the quantity of intermediate good i available at date τ, and $\varepsilon > 0$ is the elasticity of substitution across intermediate goods.

The final-good firm solves

$$\max_{\{\tilde{y}_{i,\tau}\}_{i=0}^{F}} P_\tau Y_\tau - \sum_{i=0}^{F} p_{i,\tau} \tilde{y}_{i,\tau}.$$

The associated demand for intermediate good i satisfies

$$p_{i,\tau} = P_\tau \left(\frac{Y_\tau}{\tilde{y}_{i,\tau}} \right)^{1/\varepsilon}.$$

2.3 Intermediate-Good Producers

Intermediate-good producers choose both labor input and time in production. If firm i begins production at date t and chooses duration Δ, then output delivered at date $t+\Delta$ is

$$y_{i,t}(\Delta) = \kappa^{\Delta}\left(\Delta + B_t N_i\right) A_t l_{i,t},$$

where $\kappa \in (0,1)$ is the physical cost of time-to-build, A_t is aggregate productivity, B_t scales the productivity of time-to-build, and N_i is firm-specific time-to-build productivity.

The term κ^{Δ} captures the direct cost of delaying production. The term $\Delta + B_t N_i$ captures the productivity gain from time in production. Firms with different N_i therefore choose different production durations.

Output available at date τ is the sum of goods begun in earlier periods and completed at τ:

$$\tilde{y}_{i,\tau} = \sum_{0 \le \Delta \le \tau} y_{i,\tau-\Delta}(\Delta).$$

A firm of type i that begins production at date t solves

$$\max_{\Delta, l_{i,t}} \left[p_{i,t+\Delta} \left(\prod_{j=t}^{t+\Delta} \delta_j \right) y_{i,t}(\Delta) - W_t l_{i,t} \right],$$

where δ_j is the relevant discount factor between dates.

3 Equilibrium

Definition 1 (Equilibrium)**.** *Given initial production conditions and a sequence of interest rates $\{i_t\}_{t=0}^{\infty}$, an equilibrium consists of prices*

$$\left\{ P_t, W_t, \{p_{i,t}\}_{i=0}^{F} \right\}_{t=0}^{\infty},$$

and allocations

$$\left\{ C_t, Y_t, L_t, \{l_{i,t}, \tilde{y}_{i,t}, \{y_{i,t}(t-s)\}_{s=0}^{t}, \Delta_{i,t}\}_{i=0}^{F} \right\}_{t=0}^{\infty},$$

such that:

1. *Households maximize utility given prices $\{P_t, W_t, i_t\}$.*
2. *The final-good firm maximizes profits given $\{P_t, \{p_{i,t}\}_{i=0}^{F}\}$.*

3. *Each intermediate-good firm maximizes profits given wages, discount rates, and expected future prices.*

4. *Goods and labor markets clear:*

$$C_t = Y_t,$$

$$L_t = \sum_{i=0}^{F} l_{i,t},$$

and

$$\tilde{y}_{i,\tau} = \sum_{0 \leq \Delta \leq \tau} y_{i,\tau-\Delta}(\Delta).$$

4 Steady State

In steady state, aggregate productivity, time-to-build productivity, and interest rates are constant:

$$A_t = A^{SS}, \qquad B_t = B^{SS}, \qquad i_t = i^{SS}.$$

Let δ denote the steady-state discount factor. The effective productivity of a firm choosing duration Δ is

$$z_i(\Delta) = (\kappa\delta)^{\Delta}(\Delta + BN_i)A.$$

The firm chooses Δ to maximize effective productivity. Treating Δ as continuous for the purpose of deriving intuition, the first-order condition gives

$$\Delta_i^{SS} = \max\left\{-\frac{1}{\log(\delta\kappa)} - BN_i, 0\right\}.$$

Thus, higher $\delta\kappa$ increases the desired duration of production, while firm heterogeneity in N_i generates dispersion in steady-state time-to-build choices.

The steady-state distribution of Δ_i^{SS} plays a central role in shock propagation. Firms with different production durations complete production at different horizons, so a one-time shock affects output over time through the distribution of production completion dates.

5 Small Shocks

I first consider infinitesimal one-time shocks around the steady state. Let

$$M_{k|t} = \{i : \Delta_{i,t} = k\}$$

denote the set of firms that, at date t, choose production duration k.

Proposition 1 (Local Invariance of Time-to-Build)**.** *Consider a sequence $\{i_s, A_s, B_s\}_{s\geq 0}$ such that i_s, A_s, B_s are equal to their steady-state values for all $s \neq t$, and differ from steady state only at date t. For sufficiently small deviations at date t, there exists an equilibrium in which*

$$\Delta_{i,s} = \Delta_i^{SS}$$

for all firms i and dates s.

The proposition says that small shocks do not change the endogenous time-to-build margin. Around the steady state, firms continue to choose the same production duration. Therefore, the local response of the endogenous-time-to-build economy coincides with the response of an economy in which production durations are fixed.

Corollary 1 (Output Response to a Small Productivity Shock)**.** *Suppose there is an infinitesimal one-time productivity shock at date τ. Then, for $j \geq 0$,*

$$\frac{\partial \log Y_{\tau+j}}{\partial \log A_\tau} = \frac{\eta+1}{\eta+\sigma} \frac{\sum_{i\in M_{j|\tau}} \left(z_i(j|\tau)\right)^{\varepsilon-1}}{\sum_i \left(z_i(j|s_i)\right)^{\varepsilon-1}}.$$

This expression shows that the effect of a one-time productivity shock at date τ on output at date $\tau + j$ depends on the firms whose production duration equals j. These are precisely the firms whose production begun at the shocked date is completed at horizon j.

In the special case where all effective productivities are equal, $z_i(\Delta|t) = \bar{z}$, the response simplifies to

$$\frac{\partial \log Y_{\tau+j}}{\partial \log A_\tau} = \frac{\eta+1}{\eta+\sigma} \frac{\lambda(M_{j|\tau})}{F+1},$$

where $\lambda(M_{j|\tau})$ is the number of firms completing production after j periods. Thus, the impulse response is proportional to the distribution of time-to-build durations.

6 Complementarity

The CES elasticity ε controls the degree of substitutability across intermediate goods. As ε increases, intermediate goods become closer substitutes. In this case, the firms with the highest effective productivity dominate production.

Lemma 1 (Limiting Effect of Substitutability)**.** *Suppose that there exists a unique maximum effective productivity z^* associated with firms of type i. Around the steady state:*

1. *If* $i \notin M_{k|t}$, *then*

$$\lim_{\varepsilon \to \infty} \frac{\partial \log Y_{t+k}}{\partial \log A_t} = 0.$$

2. *If* $i \in M_{k|t}$, *then*

$$\lim_{\varepsilon \to \infty} \frac{\partial \log Y_{t+k}}{\partial \log A_t} = 1.$$

The lemma shows that when intermediate goods are highly substitutable, shock propagation becomes concentrated among the firms that are most productive at the relevant horizon. In contrast, when goods are more complementary, the shock is spread more broadly across firms.

7 Growth and Time-to-Build

The model also has implications for how trend growth affects production timing. Suppose productivity grows at rate g, so that

$$A_t = (1+g)^t A_0.$$

Let Δ_i^g denote the time-to-build choice under growth rate g, and let Δ_i^{SS} denote the steady-state choice with no growth.

Lemma 2 (Growth and Production Duration)**.** *If* $g > 0$, *then*

$$\Delta_i^{SS} \geq \Delta_i^g.$$

If $g < 0$, *then*

$$\Delta_i^{SS} \leq \Delta_i^g.$$

Under positive growth, future goods are produced in a more productive environment. In equilibrium, firms perceive the relative price of later output as lower, which reduces the incentive to spend more time producing for future delivery. The optimal duration becomes

$$\Delta_i^g = \max\left\{ -\frac{1}{\log\left(\frac{\delta\kappa}{1+g}\right)} - BN_i, 0 \right\}.$$

Thus, positive growth shortens time-to-build, while negative growth lengthens it.

8 Large Anticipated Shocks

Small shocks leave the time-to-build margin unchanged. Large shocks do not. When a sufficiently large shock is anticipated, firms may alter their production timing to avoid producing for low-productivity dates.

Let $A_s = A^{SS}$ for all $s \neq t$, and let $A_t = \tilde{A} < A^{SS}$. The shock is known in advance.

Lemma 3 (Reallocation Under a Large Anticipated Shock)**.** *Suppose $B_s = B^{SS}$ and $i_s = i^{SS}$ for all s. Consider a one-time anticipated productivity decline at date t, with*

$$A_s = A^{SS} \quad \text{for } s \neq t, \qquad A_t = \tilde{A}.$$

For $\tilde{A}$ sufficiently small, firms do not produce for the shocked date:

$$y_{i,t}(\Delta_i^{SS}) = 0.$$

Moreover, adjustment shifts toward the adjacent date with the higher effective value. If

$$z_{i,t-1}(\Delta_i^{SS} + 1) > z_{i,t+1}(\Delta_i^{SS} - 1),$$

then production shifts to date $t-1$:

$$y_{i,t-1}(\Delta_i^{SS} + 1) > 0, \qquad y_{i,t+1}(\Delta_i^{SS} - 1) = 0.$$

If instead

$$z_{i,t-1}(\Delta_i^{SS} + 1) < z_{i,t+1}(\Delta_i^{SS} - 1),$$

then production shifts to date $t+1$:

$$y_{i,t-1}(\Delta_i^{SS} + 1) = 0, \qquad y_{i,t+1}(\Delta_i^{SS} - 1) > 0.$$

The lemma captures the main nonlinearity of the model. For small shocks, firms leave production timing unchanged. For large anticipated shocks, firms avoid the shocked date. The shock is therefore dampened at date t, but this dampening comes from reallocating production toward adjacent dates. Hence, large shocks can generate bunching and overshooting around the shocked date.

9 Unexpected Shocks

The response to unexpected shocks differs from the response to anticipated shocks. If the shock is unexpected, firms that have already committed to production cannot fully reallocate away from the shocked date. In particular, firms at the lowest feasible production duration are constrained and cannot reduce production time further.

Lemma 4 (Unexpected Shock)**.** *Consider a one-time unexpected productivity decline at date t:*

$$A_s = A^{SS} \quad \textit{for } s \neq t, \qquad A_t = \tilde{A}.$$

For $\tilde{A}$ sufficiently small, firms with slack in their production-duration choice delay adjustment:

$$\Delta_i^{SS} + N_i - 1 > 0 \quad \Rightarrow \quad y_{i,t}(\Delta_i^{SS}) = 0.$$

These firms instead adjust in the following period:

$$y_{i,t+1}(\Delta_i^{SS} - 1) > 0.$$

By contrast, constrained firms still produce immediately. If

$$\Delta_i^{SS} + N_i - 1 \leq 0 \quad \textit{or} \quad \Delta_i^{SS} = 0,$$

then

$$y_{i,t}(\Delta_i^{SS}) > 0.$$

The intuition is simple. Firms that can reduce their production duration respond to the unexpected shock by shifting production away from the shocked date. But firms already producing at the minimum feasible duration cannot adjust further. These constrained firms continue to produce immediately, so unexpected shocks have a stronger contemporaneous effect than anticipated shocks.

10 An Extension: Cumulative Productivity

The baseline model assumes that the productivity relevant for production is the productivity at the beginning of the production process. An alternative formulation allows productivity over the entire production period to matter. In this extension, the intermediate-good pro-

duction function becomes

$$y_{i,t}(\Delta) = \kappa^{\Delta}(\Delta + B_t N_i) l_{i,t} \sum_{p=t}^{t+\Delta} \frac{A_p}{\Delta}.$$

The market-clearing condition remains

$$\tilde{y}_{i,\tau} = \sum_{0 \leq \Delta \leq \tau} y_{i,\tau-\Delta}(\Delta).$$

In steady state, this formulation delivers the same time-to-build choice as the baseline model. Away from the steady state, however, the choice of Δ can dampen productivity shocks because firms can alter the set of productivity realizations that enter the production process.

For small shocks, the intuition is the same as in the baseline model: shocks are locally too small to change the time-to-build margin. For large shocks, future firms may take over the market from current firms, reallocating production away from dates with low productivity.

11 Monetary Policy

The time-to-build margin also changes the transmission of monetary policy. Consider a monetary authority that chooses a path of interest rates $\{i_t\}$ to minimize deviations of output from a target $\bar{Y}$:

$$\min_{\{i_t\}} \sum_t \beta^t \left(\frac{Y_t - \bar{Y}}{\bar{Y}} \right)^2.$$

In an economy without time-to-build, interest-rate changes have no real effect through the production-timing channel. With exogenous time-to-build, monetary contractions reduce output immediately because the timing of production is fixed. With endogenous time-to-build, firms can adjust production duration in response to interest-rate changes, attenuating or reshaping the output response.

Proposition 2 (Moderate Interest-Rate Increase)**.** *Suppose there is a one-time anticipated increase in the interest rate at date t. Then there exist $\underline{i}, \bar{i} > 0$ such that whenever*

$$\underline{i} < i_t < \bar{i},$$

we have

$$y_{i,t+1}(\Delta_i^{SS} - 1) > 0 \qquad \textit{and} \qquad y_{i,t}(\Delta_i^{SS}) = 0$$

for all firms i with $\Delta_i^{SS} > 1$.

The intuition is that a moderate increase in the interest rate changes the relative value of producing for different dates. Some firms reduce their time-to-build, while other firms adjust to take advantage of the resulting gap.

Proposition 3 (Large Interest-Rate Increase)**.** *Suppose there is a one-time anticipated increase in the interest rate at date t. Then there exists $\bar{i} > 0$ such that if $i_t > \bar{i}$, then production is reallocated away from periods immediately after the interest-rate increase. In particular,*

$$y_{i,j}(k) \geq 0$$

for all

$$i \in \{i : \Delta_i^{SS} > 1\}, \qquad k \in \{\underline{\Delta}_i, \ldots, \Delta_i^{SS}\}, \qquad j \in \{t+2, \ldots, t+\Delta_i^{SS}-1\},$$

and there exists at least one pair (j,k) such that

$$y_{i,j}(k) > 0.$$

If the interest rate is sufficiently high, firms do not want to produce immediately after the interest-rate increase. Instead, production is shifted toward dates after the high-interest-rate period.

12 Conclusion

This paper studies how time in production changes the propagation of macroeconomic shocks. The key mechanism is that production timing is an endogenous margin. Firms choose not only how much labor to use, but also when output will be completed. This creates a dynamic link between shocks and future output even in a model without capital.

The model has three main implications. First, small one-time shocks generate persistent effects through the steady-state distribution of time-to-build. Second, large anticipated shocks induce firms to reallocate production away from the shocked date, dampening the contemporaneous effect but shifting output across adjacent periods. Third, unexpected shocks are less easily dampened because some firms are constrained by the minimum feasible production duration.

The framework also suggests that monetary policy can interact with the production-timing margin. Interest-rate changes affect the relative payoff of producing at different

dates, and firms may respond by adjusting production duration. This creates a distinct channel through which monetary policy can affect real activity when production takes time.

Several extensions remain. A natural next step is to embed the model in a richer quantitative environment and compare the model-generated propagation of productivity and monetary shocks to empirical impulse responses. Another direction is to combine the time-to-build mechanism with production networks, allowing shocks to propagate both across firms and across time.

References